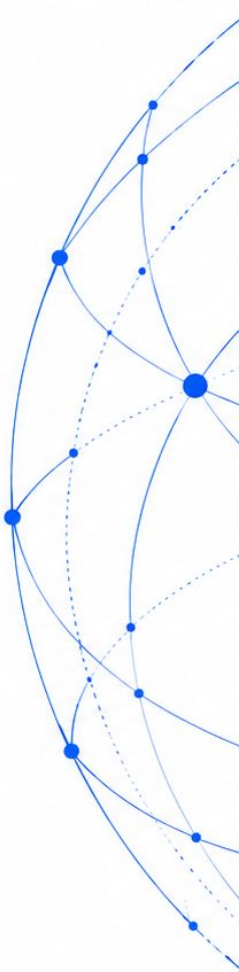


LOKA

Foundation Models on AWS Trainium

From Biology to World Models

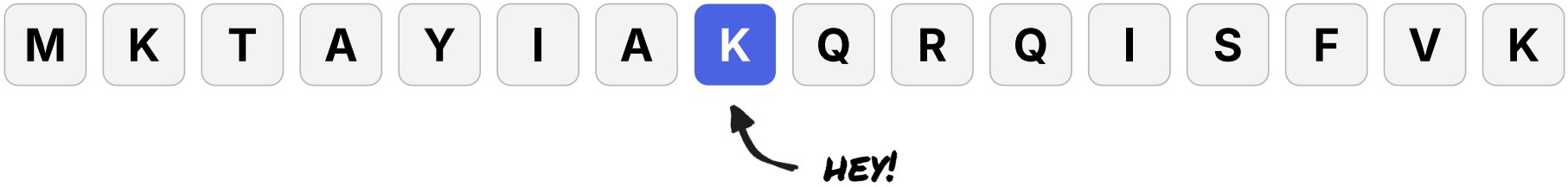
Tiago Gonçalves
Sword AI Summit 2026



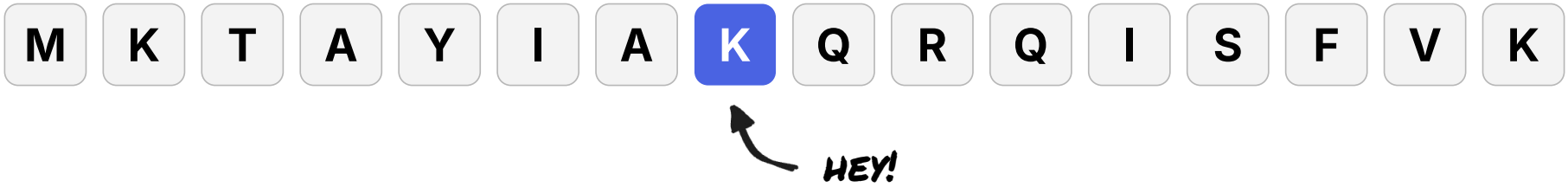
01

Proteins are code with consequences

Proteins begin as sequences



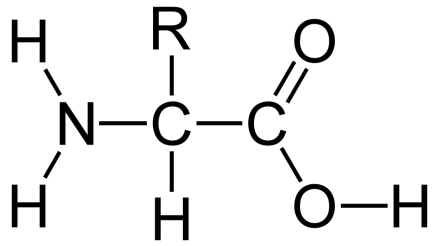
One substitution changes one letter



One substitution changes one letter

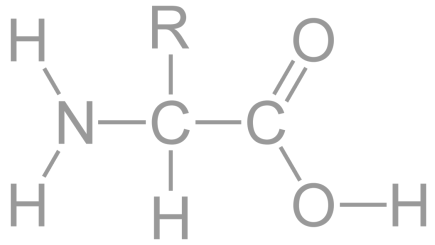
The effect can be harmless, or it can alter function

Sequence



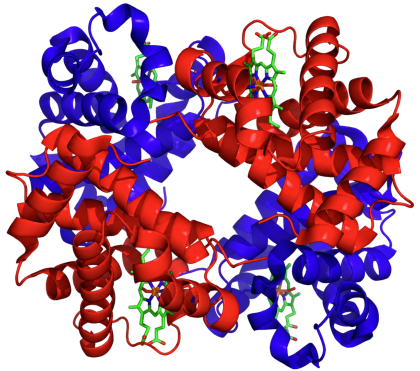
A chain built from 20 types of amino acid

Sequence



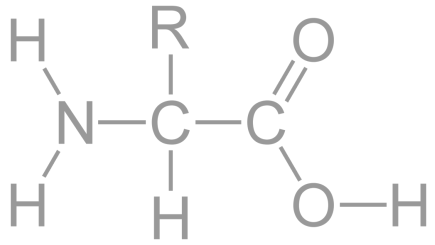
A chain built from 20 types of amino acid

Fold



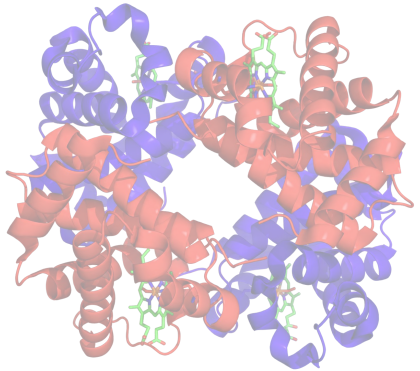
The sequence guides the chain into a 3D shape

Sequence



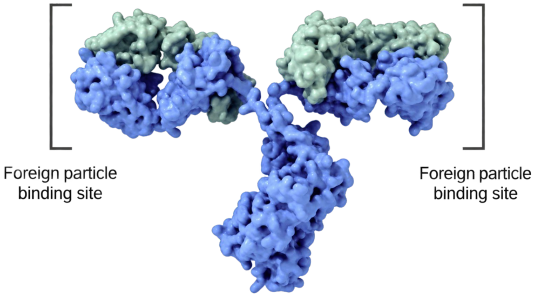
A chain built from 20 types of amino acid

Fold



The sequence guides the chain into a 3D shape

Function



That shape lets it do a job

Here, binding a foreign particle

Proteins can be altered by 1000s of single amino-acid substitutions

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Some cause disease

Proteins can be altered by 1000s of single amino-acid substitutions

Some cause disease, but most do not.



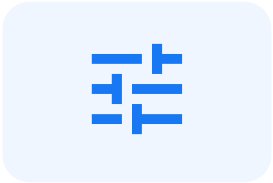
Therapeutic Target Selection

Pinpointing functionally critical protein variants to discover actionable drug targets and avoid non-viable candidate pathways



Clinical Variant Interpretation

Systematically classifying patient mutations as pathogenic or benign to resolve variants of uncertain significance



Personalized Medicine

Tailoring treatments to individual genetic profiles by predicting how specific substitutions alter therapeutic response

Traditional Approach: Slow, Bottlenecked & Expensive

- Wet-lab deep mutational scanning

Traditional Approach: Slow, Bottlenecked & Expensive

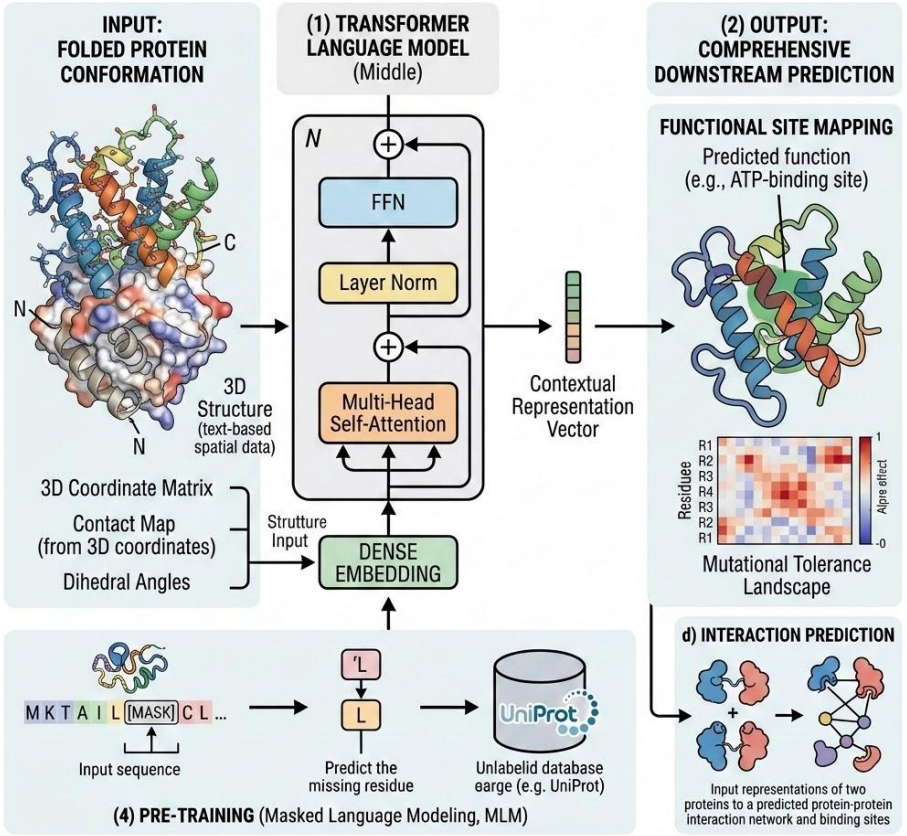
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Traditional Approach: Slow, Bottlenecked & Expensive

- Wet-lab deep mutational scanning
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- Population-scale gnomAD lookups

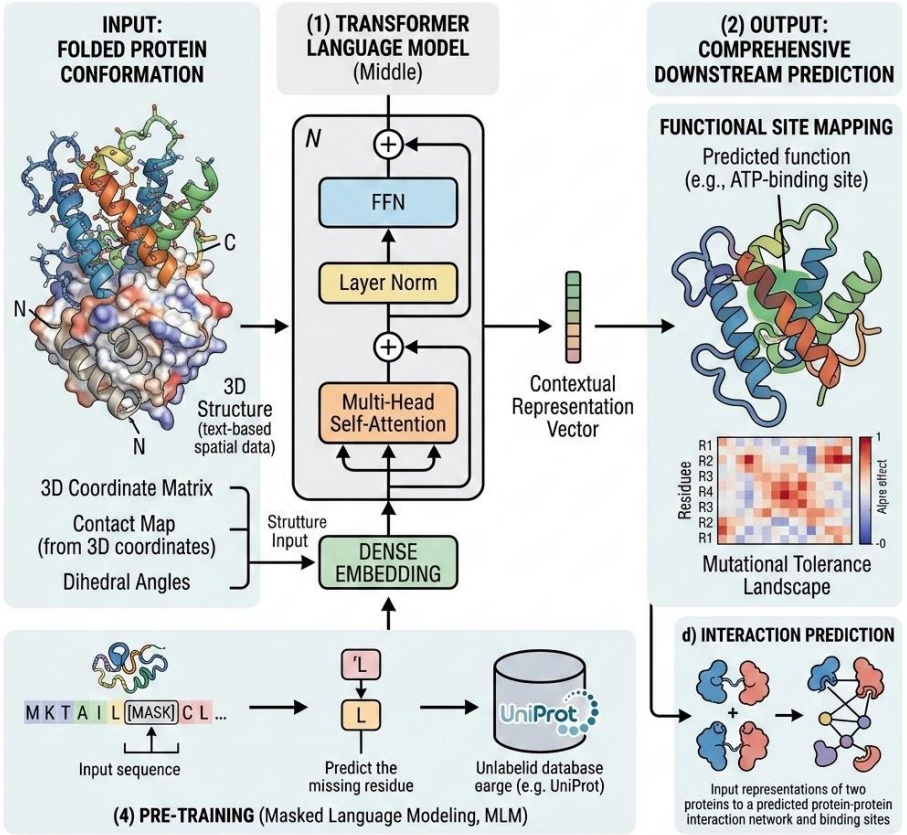
Houston, we need Protein Language Models!

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Houston, we need Protein Language Models!

Trained on hundreds of millions of natural protein sequences, learning implicitly which substitutions are tolerated at each position (and which are not)

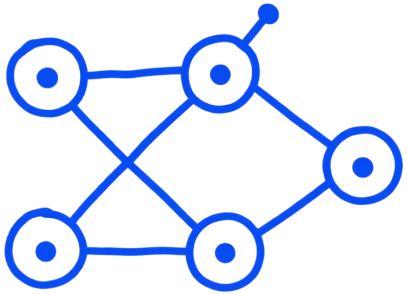


02

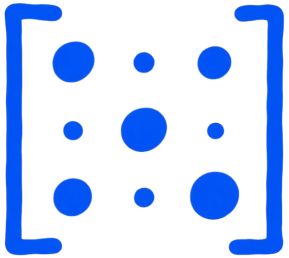
Porting Biology to Trainium2

ESMC-300M

An **open**, encoder-only protein model from **EvolutionaryScale**



300 M
Parameters



0.6 GB
Weights (bf16)

Zero-shot variant scoring: one forward pass, no fine-tuning

Trainium2? What the fuss is that?

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AWS’s chip for AI training and inference



Trainium2? What the fuss is that?

AWS's chip for AI training and inference



We ran ESMC-300M on one

TELL ME MORE...

We target the low-batch regime where per-request turnaround takes priority over hardware saturation!

1

Encoder, not decoder

ESMC does one static forward pass per sample

TELL ME MORE...

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Static shapes

Once sequence length is fixed, the compute graph doesn't change between samples

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1

Encoder, not decoder

ESMC does one static forward pass per sample

2

Static shapes

Once sequence length is fixed, the compute graph doesn't change between samples

3

trn2.3xlarge

4 logical NeuronCores, 96 GB of accelerator memory

Porting the model?

Porting the model? One simple change!

```
use_flash_attn=False
```

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It switches ESMC to standard scaled-dot-product attention: which compiles cleanly on the Neuron compiler



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Clean, without model rewrite!



Neuron SDK

Evaluated on D2Deep dataset

Evaluated on D2Deep dataset

List of Proteins

Any gene name | Any Uniprot ID | Filter

Gene	Protein	Sequence length	Total Mutations	Benign Mutations	Pathogenic Mutations	Actions
H3C2	P68431	136	2583	1626	957	View mutations
CDK4	P11802	303	5756	3605	2151	View mutations
CDKN2A	P42771	156	2963	1926	1037	View mutations
RS30	P62861	133	2526	1656	870	View mutations
RHOA	P61586	193	3666	2470	1196	View mutations
CHEK1	O14757	476	9043	5640	3403	View mutations
CTLA4	P16410	223	4236	2742	1494	View mutations
CD79B	P40259	229	4350	2649	1701	View mutations
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VHL	P40337	213	4046	2780	1266	View mutations
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EPCAM	P16422	314	5965	3951	2014	View mutations
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GATA2	P23769	480	9119	4824	4295	View mutations
CD58	P19256	250	4749	4747	2	View mutations
B2M	P61769	119	2260	1295	965	View mutations
GATA1	P15976	413	7846	3634	4212	View mutations
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Scored zero-shot scoring

The best thing?

The best thing? **Quality and compatibility!**

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490.8

variants scored / second

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variants scored / second

42.4M

variants scored / day

The best thing? **Quality and compatibility!**

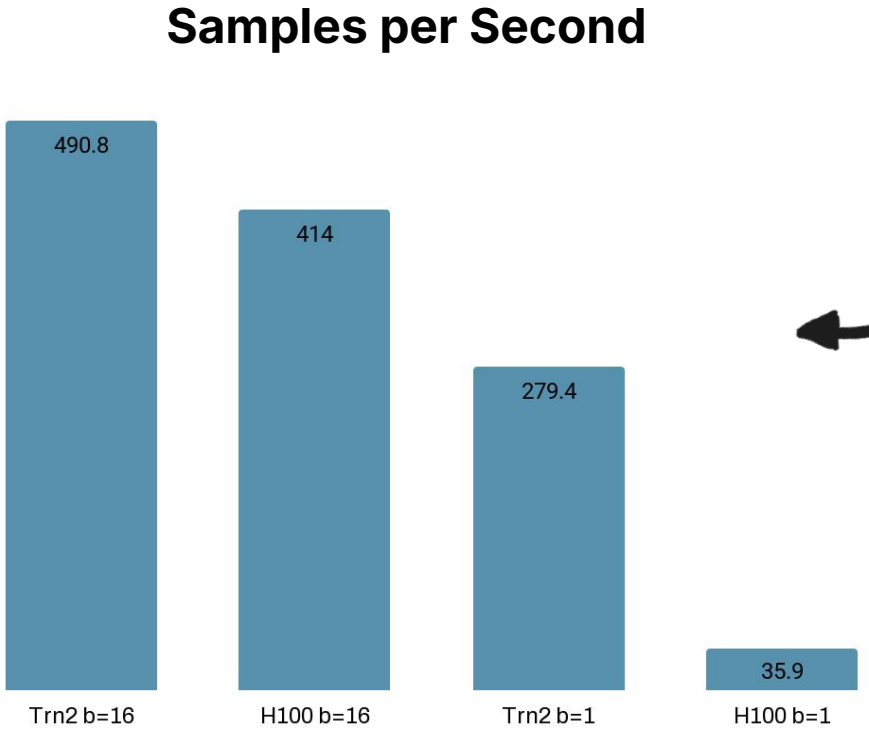
490.8	42.4M	0.8525
variants scored / second	variants scored / day	ROC-AUC on D2Deep

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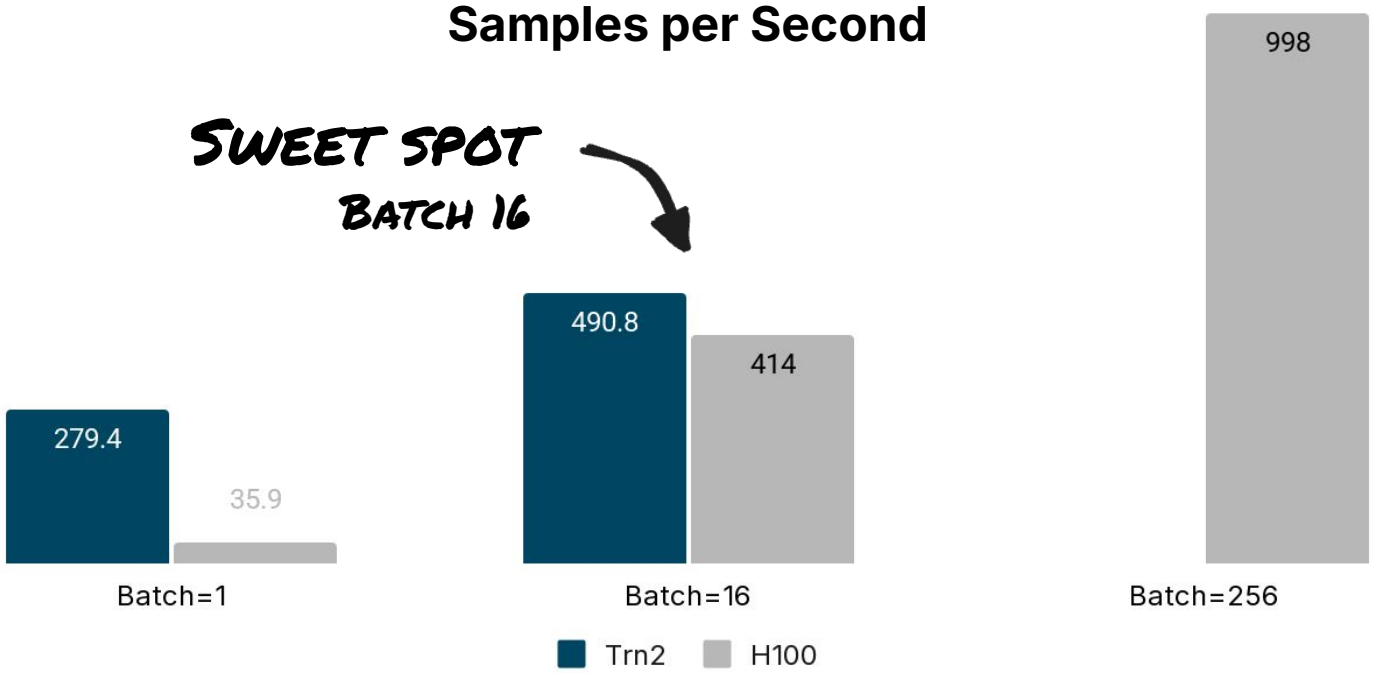
No model rewrite. No accuracy trade-off.

Full-instance throughput: +18.6%



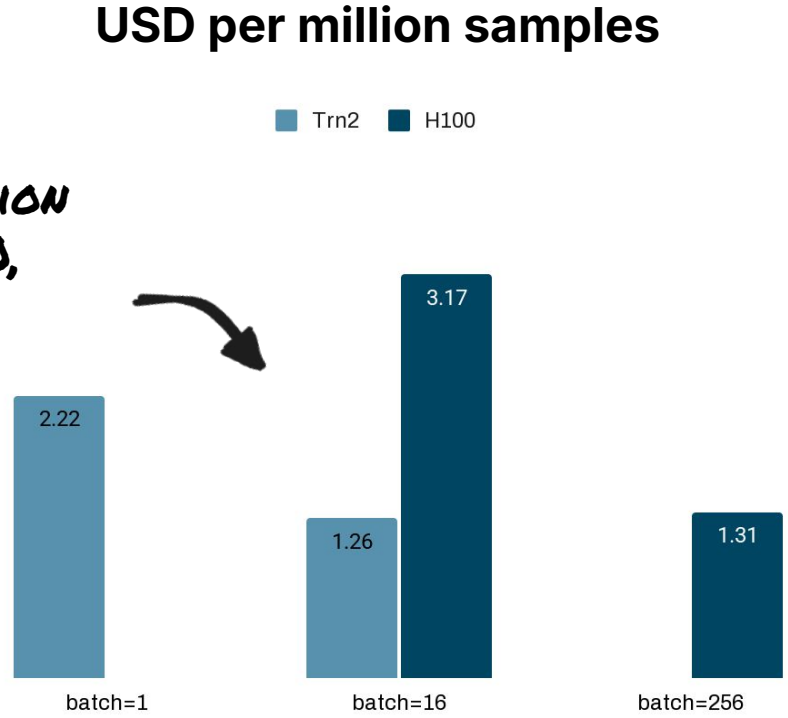
Same model, sequence length, set of batch sizes, and scoring procedure on both accelerators

Different batch-size sweet spots



Different cost per million variants scored

60% LOWER COST PER MILLION SAMPLES ON TRN2 VS. H100, AT BATCH = 16



Important outcomes for healthcare & life sciences

1

Population-scale interpretation

The infrastructure to do this at **scale**, inside a secure AWS environment, **increasingly matters**

Important outcomes for healthcare & life sciences

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Minutes, not weeks

Tens or hundreds of millions of variants are measured in days on one instance

Important outcomes for healthcare & life sciences

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Population-scale interpretation

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Minutes, not weeks

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3

Open model, predictable cost

It runs on the same accelerator infrastructure teams already operate, **predictable per-sample cost**

03

Same playbook.
New frontier.

Racing on the AWS Trainium Frontier

Racing on the AWS Trainium Frontier

30 min

single-chip training budget

Racing on the AWS Trainium Frontier

30 min

single-chip training budget

Top 10

advance to Phase 2

Racing on the AWS Trainium Frontier

30 min	Top 10	$40K
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Everything starts from a shared, nanochat-derived training pipeline

What are we cooking?

Racing on the AWS Trainium Frontier

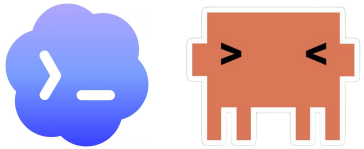
30 min	Top 10	$40K
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Everything starts from a shared, nanochat-derived training pipeline

Btw, we are in the top-10

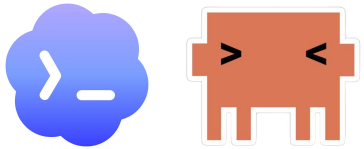
Adopting the *autoresearch* paradigm

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KUDOS TO KARPATHY

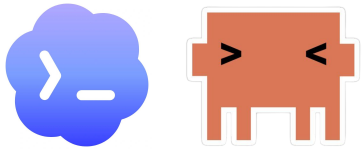
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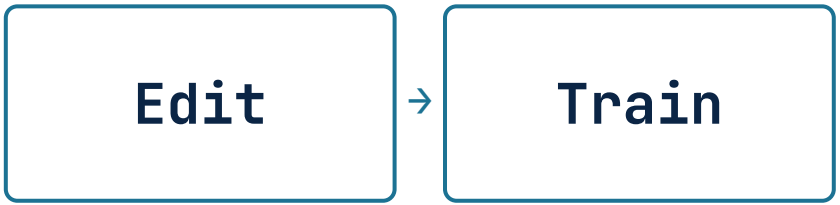
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```
Edit
```

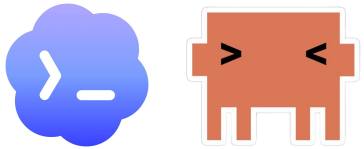
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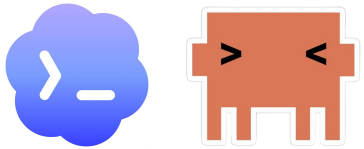
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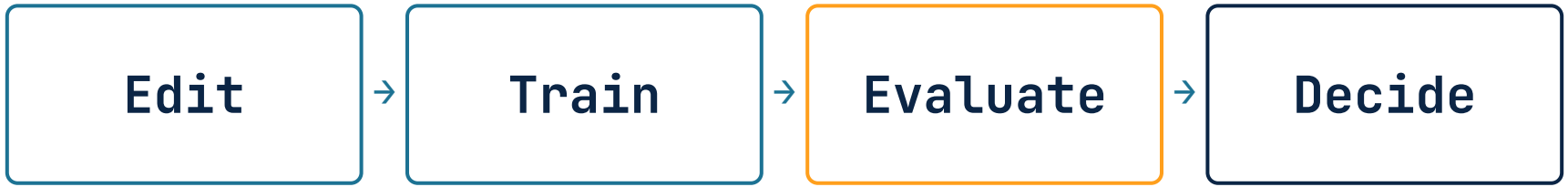
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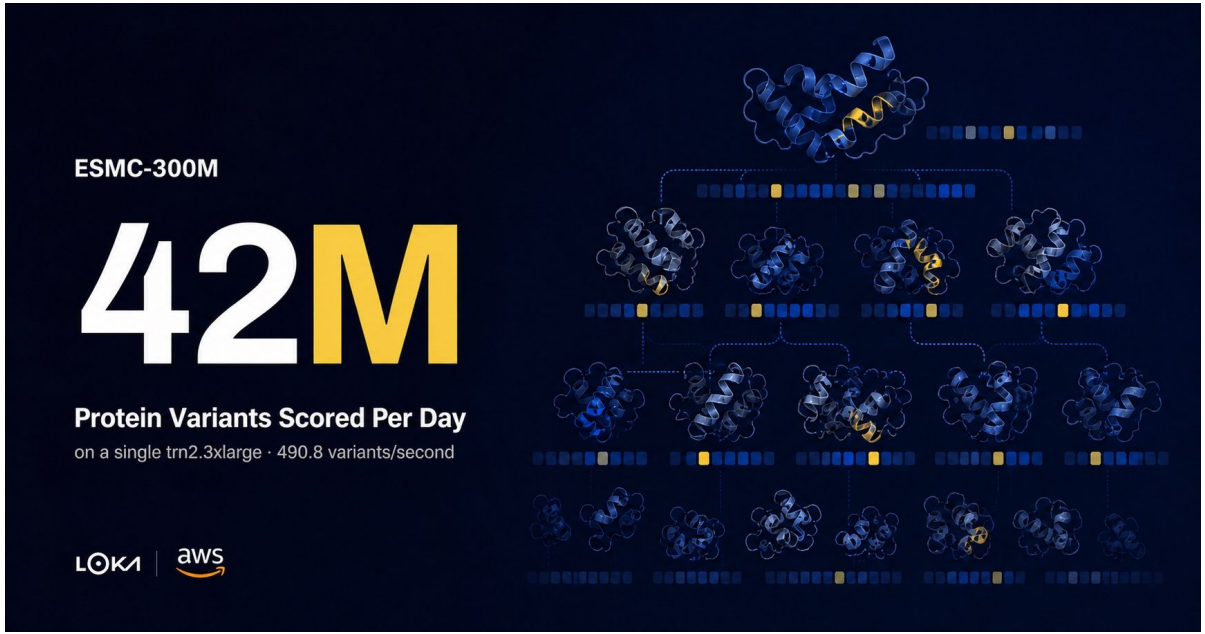
KUDOS TO KARPATHY



Playing around with World Models

Meet SANA-WM, NVIDIA NVLabs' open-source world model





LokaHQ/esmc-neuronx

https://lokahq.github.io/esmc-neuronx/

SCAN ME



LOKA

Foundation Models on AWS Trainium

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Sword AI Summit 2026

